GEOMETRY · TRANSFORMATIONS

Transformation Studio

Explore & Reflect

Name _____ Date _____ Period _____

Open **Transformation Studio** on Mr. K's Classroom: Geometry → Transformations → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict

before you touch the tool

1. What do you think happens to (2, 5) when it is reflected over the x-axis?

_____

_____

2 Explore

use the tool and record what you see

Transform △ABC with A(1, 1), B(5, 2), C(2, 5)

Transformation	Rule	A′	B′	C′
Translate 3 right, 2 down				
Reflect over the x-axis				
Reflect over y = x				
Rotate 90° counterclockwise				
Rotate 180°				
Dilate by 2				

What each transformation keeps

Transformation	Image congruent?	Orientation kept?
Translation		
Reflection		
Rotation		
Dilation (k ≠ 1)		

3 Write the rule

fill in each blank from what you observed

1. Translations, reflections and rotations are _____ motions: the image is _____ to the preimage.
2. A reflection _____ the orientation. A rotation _____ it.
3. Rotating 90° counterclockwise about the origin: (x, y) → _____.
4. Reflecting over y = x: (x, y) → _____.
5. A dilation with k ≠ 1 makes an image that is _____, not congruent.

4 Check no tool: show your work

1. Rotate (–3, 4) 180° about the origin.

2. A translation moves (2, 1) to (–1, 5). Write its rule.

3. Reflect (6, –2) over the y-axis.

4. Rotate (4, 1) 270° counterclockwise about the origin.

5 Reflect

How can you tell from the vertex order that a figure was reflected?